Higher-Order Fourier Neural Operator: Additional Experiments and Analysis

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Figure 1. **Ablation on training time, inference time and peak memory usage.** We report results for FNO with improved backbone, HO-FNO, LaMO and Transolver. The reported quatities are measured on a GPU Nvidia A100 with the configurations used in the experiments in Table 1.

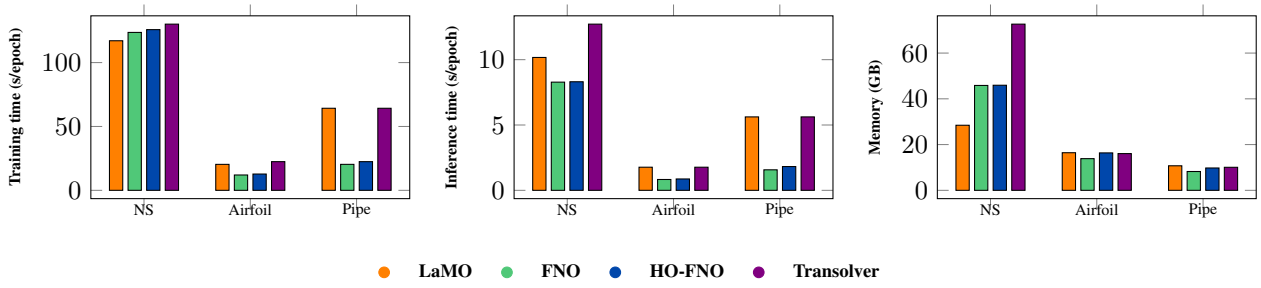
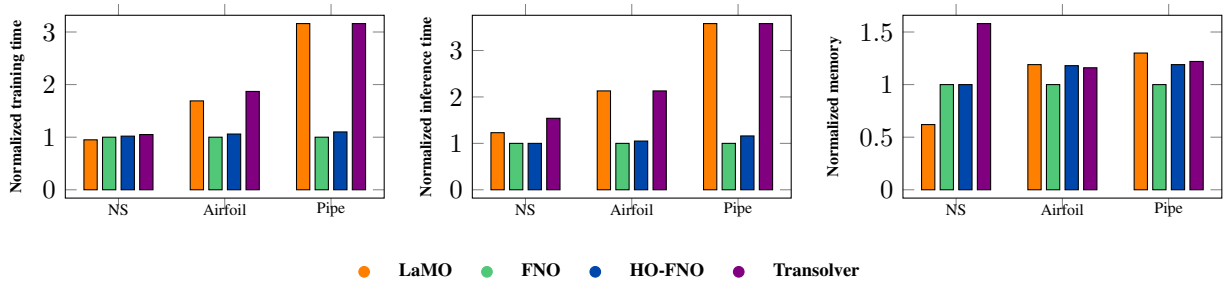


Figure 2. **Ablation on Normalized training time, inference time and peak memory usage.** We report results for FNO with improved backbone, HO-FNO, LaMO and Transolver. The reported quatities are measured on a GPU Nvidia A100 with the configurations used in the experiments in Table 1. In contrast to the previous visualization, each quantity is normalized with respect to FNO to facilitate comparison



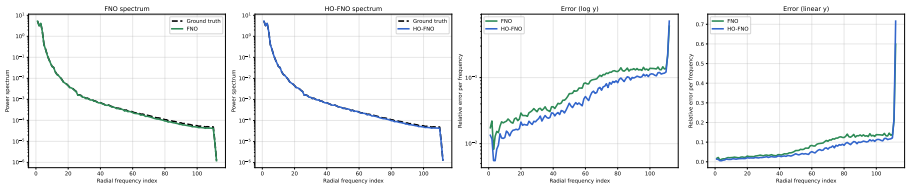


Figure 3. **Spectral Analysis on the Airfoil.** The first two plots show ground-truth and predicted energy spectra for FNO with improved backbone and HO-FNO, while the last two report frequency-wise errors. HO-FNO provides a closer match to the true spectrum and consistently lower errors across frequency bands.

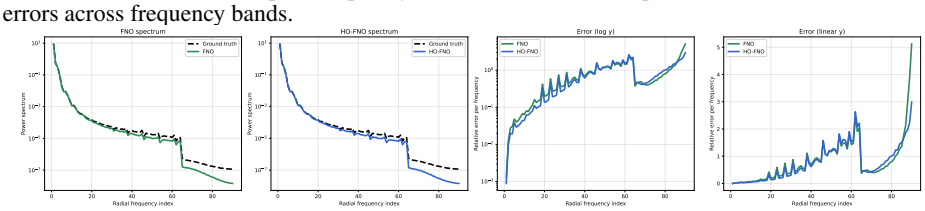


Figure 4. **Spectral Analysis on the Pipe.** The first two plots show ground-truth and predicted energy spectra for FNO with improved backbone and HO-FNO, while the last two report frequency-wise errors. HO-FNO provides a closer match to the true spectrum and reduces error across frequencies, with notable improvements at low and mid frequencies..

Table 1. Main results across standard datasets, reported as mean relative ℓ_2 error; lower is better). Baseline numbers are taken from the corresponding table in LaMO (same datasets, splits, and evaluation protocol), while **HO-FNO (Ours)** is evaluated by us under a parameter budget matched to LaMO on each task. **For the rebuttal we added FNO with improved backbone.** In the color legend, orange denotes the best baseline, blue the second-best, and violet the third-best. Best result for each dataset is **in bold**.

	Operator	Res. equiv.	Regular Grid		Structured Mesh			Point Cloud
			Navier–Stokes	Darcy	Plasticity	Airfoil	Pipe	Elasticity
Classic	UNET	✗	0.1982	0.0080	0.0051	0.0079	0.0065	0.0235
	Resnet	✗	0.2753	0.0587	0.0233	0.0391	0.0120	0.0262
	SWIN	✗	0.2248	0.0397	0.0170	0.0270	0.0109	0.0283
	DeepONet	✗	0.2972	0.0588	0.0135	0.0385	0.0097	F
Transformer	Galerkin	✗	0.1401	0.0084	0.0120	0.0118	0.0098	0.0240
	HT-Net	✗	0.1847	0.0079	0.0333	0.0065	0.0059	/
	OFormer	✗	0.1705	0.0124	0.0017	0.0183	0.0168	0.0183
	GNOT	✗	0.1380	0.0105	0.0336	0.0076	0.0047	0.0086
	FactFormer	✗	0.1214	0.0109	0.0312	0.0071	0.0060	/
	ONO	✓	0.1195	0.0076	0.0048	0.0061	0.0052	0.0118
	Transolver	✗	0.0957	0.0059	0.0013	0.0053	0.0046	0.0064
SSM	LaMO	✗	0.0460	**0.0039**	0.0007	**0.0041**	**0.0038**	**0.0050**
Frequency	WMT	✗	0.1541	0.0082	0.0076	0.0075	0.0077	0.0359
	U-FNO	✓	0.2231	0.0183	0.0039	0.0269	0.0056	0.0239
	FNO	✓	0.1556	0.0108	0.0074	0.0138	0.0067	0.0229
	U-NO	✓	0.1713	0.0113	0.0034	0.0078	0.0100	0.0258
	F-FNO	✓	0.2322	0.0077	0.0047	0.0078	0.0070	0.0263
	LSM	✗	0.1535	0.0065	0.0025	0.0059	0.0050	0.0218
	FNO with improved backbone (Ours)	✓	0.0976	0.0068	0.0066	0.0065	0.0069	0.205
	HO-FNO (Ours)	✓	**0.0449**	0.0052	**0.0006**	0.0051	0.0052	0.0199
	Increment over FNO with improved backbone		**54.1%**	**21.2%**	**90.8%**	**26.1%**	**24.6%**	**2.9%**
	Increment over the best Frequency model		**70.7%**	**20.0%**	**76.0%**	**13.6%**	−4%	**8.7%**
	Increment over the best Transformer model		**53.1%**	**2%**	**53.8%**	**3.8%**	−13.0%	−211%
	Increment over the best Res. Equiv. model		**63.4%**	**46.2%**	**76.0%**	**82.4%**	**0.0%**	−68.6%

Table 2. **Ablation on the structure of the matrices A_s on the Darcy flow equation for HO-FNO of order 2.**

Structure	loss
Dense	5.2×10^{-3}
identity	7.4×10^{-3}
diagonal	7.0×10^{-3}
low-rank (128)	6.6×10^{-3}
low-rank (64)	7.5×10^{-3}
diagonal + low-rank (64)	7.2×10^{-3}
diagonal + low-rank (128)	6.4×10^{-3}
shared	6.7×10^{-3}

Table 3. **Ablation on training time, inference time and peak memory usage.** We report results for FNO with improved backbone, HO-FNO, LaMO and Transolver. The reported quatities are measured on a GPU Nvidia A100 with the configurations used in the experiments in Table 1. For each metric, we report the absolute value and the relative variation (in %) with respect to FNO. Negative values indicate an improvement over FNO (lower is better), while positive values indicate a degradation.

Dataset	Model	Training time (s)		Inference time (s)		Memory (GB)	
		Value	Δ (%)	Value	Δ (%)	Value	Δ (%)
NS	LaMO	117.10	-5.3	10.17	+22.8	28.45	-38.0
	FNO	123.63	0.0	8.28	0.0	45.85	0.0
	HO-FNO	125.81	+1.8	8.31	+0.4	45.94	+0.2
	Transolver	130.11	+5.2	12.72	+53.6	72.65	+58.5
Airfoil	LaMO	20.33	+69.2	1.77	+113.3	16.44	+18.7
	FNO	12.01	0.0	0.83	0.0	13.85	0.0
	HO-FNO	12.76	+6.2	0.87	+4.8	16.36	+18.1
	Transolver	22.45	+86.9	1.77	+113.3	16.06	+16.0
Pipe	LaMO	64.25	+215.9	5.62	+258.0	10.73	+30.4
	FNO	20.34	0.0	1.57	0.0	8.23	0.0
	HO-FNO	22.45	+10.4	1.82	+15.9	9.77	+18.7
	Transolver	64.25	+215.9	5.62	+258.0	10.07	+22.4

Table 4. **Relative L^2 test errors for FNO and HO-FNO across Navier–Stokes, Airfoil, and Pipe datasets.**Full denotes the standard loss used in Table 1, while Low, Medium, and High correspond to losses computed on disjoint frequency bands of the solution. HO-FNO consistently improves over FNO across all frequency regimes.

	Navier-Stokes				Airfoil				Pipe			
	Full	Low	Medium	High	Full	Low	Medium	High	Full	Low	Medium	High
FNO	0.0976	0.145982	0.282080	0.458579	0.0065	0.025199	0.070703	0.146905	0.0069	0.191112	1.040052	1.253831
HO-FNO	0.0449	0.055541	0.206717	0.372876	0.0051	0.017370	0.045756	0.121412	0.0054	0.141260	1.031456	1.182214
Improvement over FNO	54.10%	61.95%	26.72%	18.69%	26.1%	31.07%	35.28%	17.35%	32.7%	26.1%	0.83%	5.71%

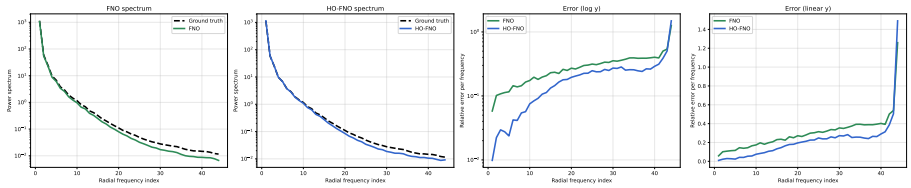


Figure 5. **Spectral Analysis on Navier–Stokes.** The first two plots show ground-truth and predicted energy spectra for FNO with improved backbone and HO-FNO, while the last two report frequency-wise errors. HO-FNO provides a closer match to the true spectrum and consistently lower errors across frequency bands.

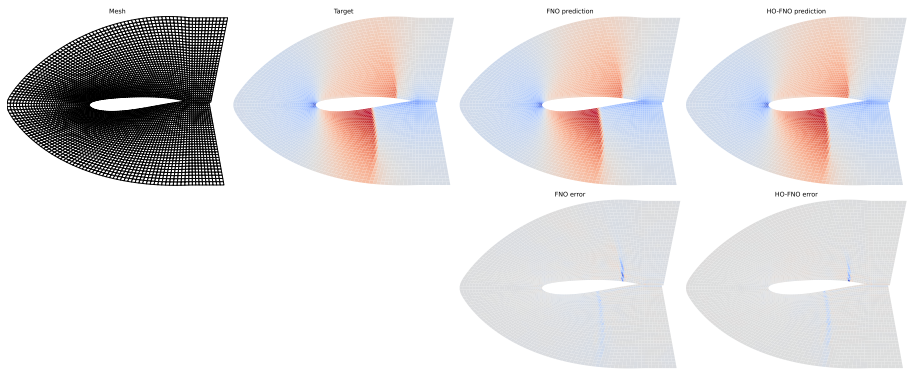


Figure 6. **Qualitative comparison on the Airfoil dataset.** From left to right: computational mesh, ground truth, FNO prediction, and HO-FNO prediction. Bottom: corresponding error maps. HO-FNO better captures flow structures around the airfoil and yields reduced errors compared to FNO with improved backbone.

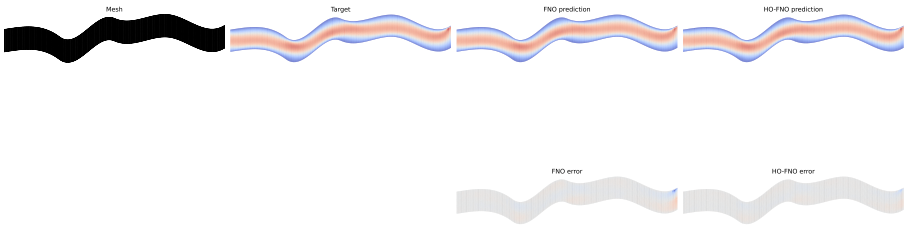


Figure 7. **Qualitative comparison on the Pipe dataset.** From left to right: computational mesh, ground truth, FNO prediction, and HO-FNO prediction. Bottom: corresponding error maps. HO-FNO more accurately captures the flow along the curved geometry and reduces errors compared to FNO with improved backbone.

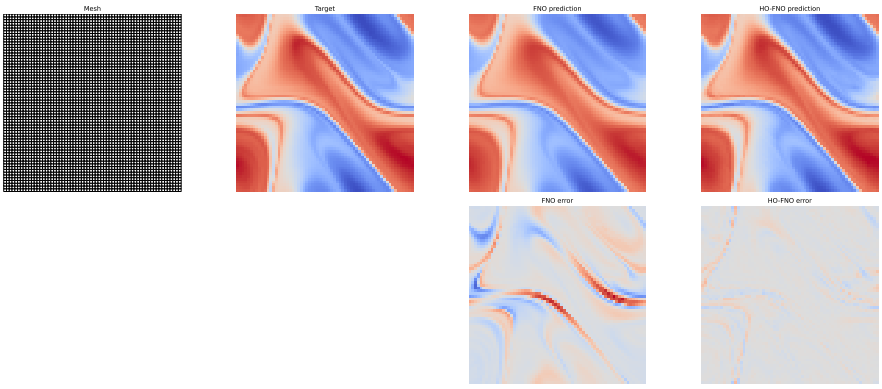


Figure 8. **Qualitative comparison on the Navier–Stokes dataset.** From left to right: input/mesh, ground truth, FNO prediction, and HO-FNO prediction. Bottom: corresponding error maps. HO-FNO better captures fine-scale flow structures and reduces errors compared to FNO with improved backbone.